Journal Upload – Build, Deploy and Configuration Guide

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Overview

Journal Upload is one SAP CAP application deployed to Cloud Foundry as a single MTA (`zmaxx-f2548-ext`). It reads master data from and posts journals to S/4HANA system **S43** through the F2548 posting service, and keeps every uploaded file in SAP Document Management Service (DMS).

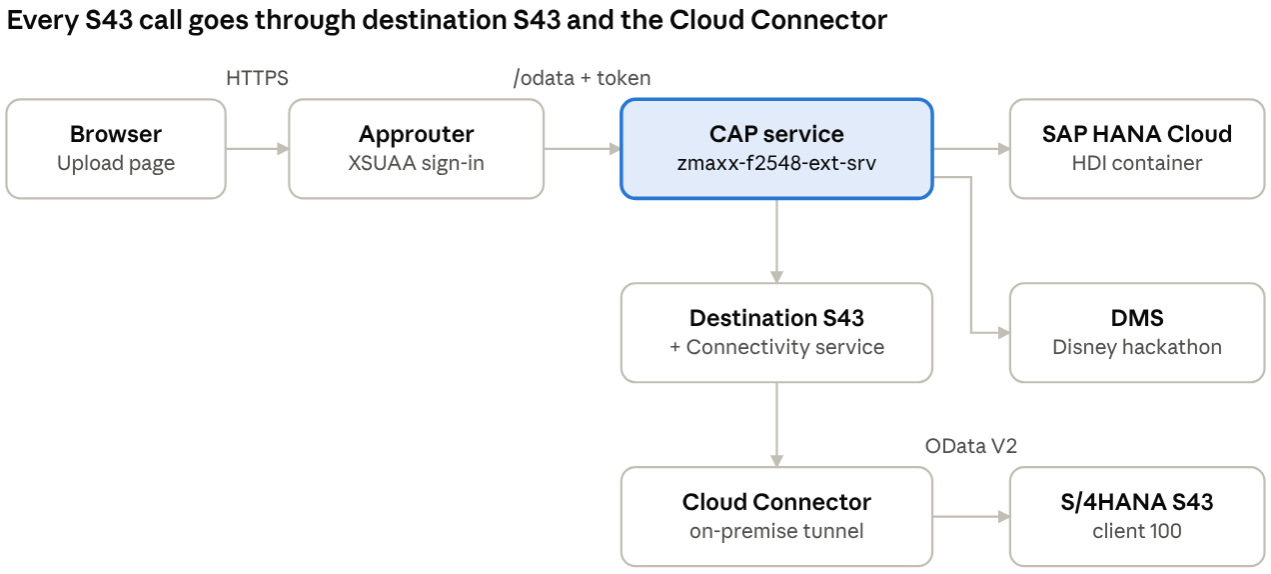
Item	Value
BTP region	US10 (Cloud Foundry API `https://api.cf.us10-001.hana.ondemand.com`)
Org / space	`poc-mc10-org` / `poc-mc-space10`
App URL	https://poc-mc10-org-poc-mc-space10-zmaxx-f2548-ext.cfapps.us10-001.hana.ondemand.com
Git repository	Azure DevOps, "Disney Hackathon"
Project folder	`zmaxx-f2548-ext` (CAP Node.js, `@sap/cds` 10, SAPUI5 1.120.21)

What the MTA creates:

MTA name	Kind	Purpose
`zmaxx-f2548-ext`	Approuter app	Serves the upload page, the samples and the Excel add-in; handles sign-in (XSUAA) and forwards `/odata/*` to the service
`zmaxx-f2548-ext-srv`	Node.js app	CAP service: upload, validation, split, posting to S43, DMS storage
`zmaxx-f2548-ext-db-deployer`	HDI deployer (runs once per deploy)	Creates and updates tables and loads the code lists in HANA

MTA name	Kind	Purpose
`zmaxx-f2548-ext-db`	HANA HDI container (`hdi-shared`)	Uploads, lines, journals, messages, TTYPE rules, split settings
`zmaxx-f2548-ext-auth`	XSUAA (`application`)	Company sign-in; no roles (open access)
`zmaxx-f2548-ext-destination`	Destination service (`lite`)	Looks up destination `S43`
`zmaxx-f2548-ext-connectivity`	Connectivity service (`lite`)	Cloud Connector tunnel to the on-premise S43

DMS is **not** part of the MTA: it is an existing service instance whose credentials are given to the service app as environment variables (see Document Management Service).



architecture · BTP services and S43

The browser only talks to the approuter; the CAP service holds every connection to HANA, DMS and, through the destination and Cloud Connector, S43.

Prerequisites

Build and deploy from SAP Business Application Studio (BAS); the Full Stack Cloud Application dev space already has every tool below.

Need	Detail
Tools	Node.js 20 or later, `npm`, Cloud MTA Build Tool (`mbt`), Cloud Foundry CLI (`cf`) with the MultiApps plugin (`cf deploy`), `@sap/cds-dk`
Cloud Foundry login	`cf login -a https://api.cf.us10-001.hana.ondemand.com --sso`, then org `poc-mc10-org`, space `poc-mc-space10`. The login expires after some hours; log in again when `cf` or `cds bind` reports an expired token
Space role	Space Developer in `poc-mc-space10`
Entitlements in the subaccount	SAP HANA Cloud (`hdi-shared`), Authorization and Trust Management (`application`), Destination (`lite`), Connectivity (`lite`), Document Management Service, Integration Option
SAP HANA Cloud	A running HANA Cloud database in the space or shared to it; if it is stopped, start it in HANA Cloud Central before deploying
Cloud Connector	Connected to the subaccount and exposing S43 (see Destination S43)
DMS	A DMS service instance with a service key and a repository (see Document Management Service)

Build and deploy

One build command and one deploy command put the whole application on BTP; after the first deploy, the DMS variables must be set once on the service app.

1. Get the latest code and install packages:

```
cd ~/projects/zmaxx-f2548-ext
git pull
npm ci
```

2. Check the switches in `package.json` under `cds.zmaxx.s43` before building, especially `allowPosting` (see Environment variables and settings). The deployed app uses the same values.
3. Build the archive:

```
mbt build
```

This runs `npx cds build --production` (writes `gen/srv` and `gen/db`), copies the UI to `app/router/resources` and writes `mta_archives/zmaxx-f2548-ext_1.0.0.mtar`.

4. Log in to Cloud Foundry (see Prerequisites) and deploy:

   ```
   cf deploy mta_archives/zmaxx-f2548-ext_1.0.0.mtar
   ```

   It creates or updates the service instances, runs the HDI deployer, then deploys the service app and the approuter.
5. First deploy only, or if the DMS variables are missing: set them on the service app and restage (commands in Document Management Service).
6. Check the result:

   ```
   cf apps
   cf logs zmaxx-f2548-ext-srv --recent
   ```

   Both apps must show `started`. Open the app URL, sign in with your BTP user and check that the upload page loads, Recent Uploads fills and Files in DMS lists the folder.

Build rules to keep. `mbt` runs each `before-all` command without a shell, so `mta.yaml` must keep one command per line, with no `&&` or pipes. The HDI deployer reloads the code-list CSV files in `db/data` on every deploy, so deploy only from a branch that has the latest code: an older branch removes newer status values such as Partly Valid from the shared database.

Redeploying only the code uses the same two commands; existing service instances and their data are kept. To stop and start without deploying: `cf restart zmaxx-f2548-ext-srv`.

Run locally in Business Application Studio

Locally the app runs in BAS against the real HANA container, the real S43 and the real DMS; only sign-in is switched off.

1. Log in to Cloud Foundry (see Prerequisites). `cds bind` needs it to fetch the HANA credentials.
2. Bind the HANA container once per workspace (writes `.cdsrc-private.json`, not committed):

   ```
   cds bind db --to zmaxx-f2548-ext-db:zmaxx-f2548-ext-db-key --profile
   development
   ```

3. Create `.env` in the project root with the DMS variables (see Environment variables and settings). It is listed in `.gitignore`; never commit it.
4. Start the server:

   ```
   npm run watch
   ```

   This is `cds watch --profile development`. It serves on port 4004; BAS shows the link `https://port4004-<workspace>.applicationstudio.cloud.sap/upload/webapp/index.html`.
5. To deploy table changes to HANA from BAS without a full MTA deploy:

   ```
   cds deploy --to hana --profile development
   ```

In the `development` profile S43 is reached through the BAS destination proxy (`devProxy` `127.0.0.1:8887`, URL `http://S43.dest`), so destination S43 must exist in the subaccount the dev space belongs to. `cds watch` restarts on every saved file in `srv`, `db` or `app`, which also stops a validation running in the background.

Environment variables and settings

The app needs six DMS environment variables; everything else comes from service bindings or from the `cds` section of `package.json`.

DMS environment variables. Locally they live in `.env` in the project root (git-ignored). On BTP they are set on the app `zmaxx-f2548-ext-srv` with `cf set-env`. In the `hybrid` profile the local server reads them from that BTP app instead (`cds.zmaxx.dms.cfApp`).

Variable	Value	Where it comes from	Secret
`DMS_BASE_URL`	`https://api-sdm-di.cfapps.us10.hana.ondemand.com`	DMS service key, `uri` (no trailing slash)	No
`DMS_TOKEN_URL`	`https://enerbridge-cpb.authentication.us10.hana.ondemand.com/oauth/token`	DMS service key, `uaa.url` + `/oauth/token`	No
`DMS_CLIENT_ID`	(from the key)	DMS service key, `uaa.clientid`	Yes
`DMS_CLIENT_SECRET`	(from the key)	DMS service key, `uaa.clientsecret`	Yes
`DMS_DEFAULT_REPOSITORY_ID`	`68c06418-89f7-4654-8d97-fb898003abc4`	DMS repository ID (see Document Management Service)	No
`DMS_DEFAULT_FOLDER_PATH`	`Disney hackathon`	Folder under the repository root; created on first upload. Defaults to `Journal uploads` if not set	No

Never put the client ID or secret in Git, in this document or in chat; take them from the service key when needed.

App switches in `package.json` (`cds.zmaxx`). They are built into the deployed app, so change them before `mbt build`.

Setting	Current value	Meaning
`s43.destination`	`S43`	Name of the BTP destination to S/4HANA
`s43.sapCheck`	`true`	Run SAP's own simulation of every error-free journal during validation
`s43.allowPosting`	`true`	Allow posting and parking in S43. `false` = validate only; Post Validated Journals is disabled
`s43.devProxy` (development profile)	`127.0.0.1:8887`	BAS destination proxy, local only
`dms.cfApp` (hybrid profile)	`zmaxx-f2548-ext-srv`	BTP app to read the DMS variables from
`ai.enabled` / `ai.model`	`false` / `gpt-4o-mini`	Use SAP Generative AI Hub for error explanations; off = built-in texts
`server.body_parser.limit`	`40mb`	Largest upload request (files up to 25 MB)

Profiles.

Profile	Used by	Sign-in	Database
`development`	`npm run watch` in BAS	None (`dummy`: everyone has full access)	HANA via `cds bind`
`hybrid`	`cds watch --profile hybrid`	CAP mocked users (no role checks in the app)	HANA via `cds bind`, DMS variables from the BTP app
`production`	The deployed app	XSUAA (BTP company sign-in, no roles)	HANA HDI container bound by the MTA

`allowPosting` is currently `true`, so the deployed app can create real journals in S43; set it to `false` before building if the BTP app should only validate.

Destination S43

All S/4HANA access goes through one subaccount destination named `S43`, which points through the Cloud Connector to S43 client 100; the app has no other S/4 connection.

Destination settings (BTP cockpit → Connectivity → Destinations):

Property	Value
Name	`S43` (must match `cds.zmaxx.s43.destination`)
Type	`HTTP`
Proxy Type	`OnPremise`
URL	`http://<virtual host>:<virtual port>` as exposed in the Cloud Connector
Authentication	Typically `BasicAuthentication` with a technical S43 user; confirm in the cockpit
Additional property `sap-client`	`100`
Additional property `WebIDEEnabled`	`true` (needed for the BAS proxy used locally)
Additional property `WebIDEUsage`	`odata_gen` (recommended for BAS, not read by the app)
Additional property `HTML5.DynamicDestination`	`true` (recommended, not read by the app)

Open question: the exact virtual host, port and technical user of S43 are not recorded in the project; read them from the destination in the cockpit and fill them in here.

Cloud Connector. It must be connected to this subaccount and expose the S43 virtual host with access to path prefix `/sap/opu/odata/sap/` (or the two service paths below). Journals created through the app show this technical user as the creator in S43 (FB03, document header).

S43 services the app calls:

Service	Used for
`FAC_GL_DOCUMENT_POST_SRV` (the F2548 service)	Master data value helps (company code, G/L account, cost center, business area, document type, currency, tax code, Transaction Type); `FinsPostingGLHeaders` and `$batch` to build the journal; `Simulate` for the SAP check; `Post` and `ParkAsComplete`; `IsWorkflowActive`
`API_CV_ATTACHMENT_SRV`	Attaching the uploaded file to the created journal (business object `BKPF`)

Both services must be active in S43 (`/IWFND/MAINT_SERVICE`) and the technical user needs the authorizations of a General Ledger Accountant for company code 1710 (posting, parking, simulation) plus attachment authorizations.

How the app connects. Deployed on BTP, it reads destination `S43` through the destination and connectivity services bound by the MTA. In BAS (`development` profile) it calls `http://S43.dest` through the BAS proxy `127.0.0.1:8887`. Reads retry up to 3 times and the SAP check up to 5 times when S43 times out (HTTP 504).

Document Management Service

Every uploaded file is stored in DMS repository `68c06418-89f7-4654-8d97-fb898003abc4`, folder `Disney hackathon`; HANA keeps only the file's metadata and its DMS object ID.

Item	Value
Service	SAP Document Management Service, Integration Option
API endpoint	`https://api-sdm-di.cfapps.us10.hana.ondemand.com` (CMIS browser binding: `<endpoint>/browser/<repository ID>/root`)
Token endpoint	`https://enerbridge-cpb.authentication.us10.hana.ondemand.com/oauth/token` (client credentials)
Repository ID	`68c06418-89f7-4654-8d97-fb898003abc4`

Item	Value
Folder	`Disney hackathon`, one level under the repository root; created automatically on the first upload
File name in DMS	Upload time stamp + original name, e.g. `2026-10-23-11-46-17 F2548_Errors_Set2_Format.csv`
Stored in HANA	`Uploads.dmsObjectId`, `dmsFileName`, `dmsStatus`; `SourceFiles` holds repository, folder, object ID and file hash

Open question: the name of the DMS service instance and its service key are not recorded in the project; add them here.

Setting it up for the deployed app. The MTA does not bind DMS, so the service app gets the credentials as environment variables. Run once after the first deploy, and again whenever the service key changes:

```
cf set-env zmaxx-f2548-ext-srv DMS_BASE_URL "https://api-sdm-di.cfapps.us10.hana.ondemand.com"
cf set-env zmaxx-f2548-ext-srv DMS_TOKEN_URL "https://enerbridge-cpb.authentication.us10.hana.ondemand.com/oauth/token"
cf set-env zmaxx-f2548-ext-srv DMS_CLIENT_ID "<uaa.clientid from the service key>"
cf set-env zmaxx-f2548-ext-srv DMS_CLIENT_SECRET "<uaa.clientsecret from the service key>"
cf set-env zmaxx-f2548-ext-srv DMS_DEFAULT_REPOSITORY_ID "68c06418-89f7-4654-8d97-fb898003abc4"
cf set-env zmaxx-f2548-ext-srv DMS_DEFAULT_FOLDER_PATH "Disney hackathon"
cf restage zmaxx-f2548-ext-srv
```

Check with `cf env zmaxx-f2548-ext-srv` (section User-Provided) after each `cf deploy`, as a redeploy may reset them. If the repository does not exist yet, create it with the DMS admin API or the DMS administration app (repository type internal) and put its ID in `DMS_DEFAULT_REPOSITORY_ID`.

What the app does with DMS:

- Upload: the file goes to the folder before validation starts; if DMS fails, the upload still runs and shows the warning "could not be stored in Document Management".
- Files in DMS view: lists the folder (newest first, up to 200) with Download and Open Results.
- Download Lines with Errors: reads the original file back from DMS to copy the rows as uploaded.

- Posting: reads the file from DMS and attaches it to the SAP journal. This currently fails because of a key-format bug (see Troubleshooting).

Secrets. The client secret was once shared in a chat; regenerate the service key and update `DMS_CLIENT_SECRET` in `.env` and on the BTP app.

Troubleshooting

Symptom	Cause	Fix
`cf` or `cds bind` says "The token expired, was revoked"	Cloud Foundry login expired	`cf login -a https://api.cf.us10-001.hana.ondemand.com --sso`
`mbt build` fails on a `before-all` command	A command with `&&` or a pipe in `mta.yaml`	One command per line
HANA error "invalid column name" after a deploy	HANA has an older or newer table layout than the code	Deploy the same branch to HANA: `cf deploy` or `cds deploy --to hana --profile development`
Status badges show only a blue icon, no name	Status values such as `PARTIAL` were removed by a deploy from an older branch	Deploy the latest code; the service also restores the two partial statuses on start
Many "504 Gateway Timeout" from S43	S43 or the BAS proxy is slow	The app retries; if a journal is held with "SAP's check could not run", upload it again later
"There is no item category assigned to account 164500" (GLT2/076)	G/L 164500 is not set up for posting in S43	Use accounts that S43 accepts, e.g. 11001020 / 11001030
Upload shows "could not be stored in Document Management"	DMS variables missing or wrong, or secret expired	Check `cf env zmaxx-f2548-ext-srv`, set the variables, `cf restage`
Journal posted, warning "source file could not be attached … journal entry doesn't exist"	Known bug: the attachment key uses the document number without leading zeros	Fix in `srv/lib/s4-client.js` (`objectKey`, pad the number to 10 digits), then redeploy
Post Validated Journals is greyed out	`allowPosting` is `false` in the deployed build	Set it to `true` in `package.json`, rebuild and redeploy

Symptom	Cause	Fix
Local server stops validations halfway	`cds watch` restarted after a file was saved	Do not save project files while a run is in progress